

Skill Cortex V1

Business Requirements Document (BRD)

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Project: Skill Cortex

Release: Version 1

Document Type: Business Requirements Document

1. Executive Summary

Skill Cortex is a web-based webinar and course booking platform designed to allow users to discover educational webinars, select their department, choose a suitable course and time slot, make online payments, and receive automated notifications and reminders.

The platform will provide two primary interfaces:

- **User Portal** — for discovering webinars, booking slots, making payments, and receiving notifications.
- **Admin Portal** — for managing departments, webinars, slots, users, bookings, payments, and notifications.

Version 1 will focus on the complete webinar booking and payment lifecycle. AI-powered recommendations and advanced skill intelligence are outside the scope of V1.

2. Business Objective

The primary objective of Skill Cortex V1 is to provide a centralized platform for managing paid educational webinars and courses.

The system should:

1. Provide an attractive landing page for discovering available courses.
 2. Allow users to securely register and log in.
 3. Allow users to select their department.
 4. Display relevant webinars and courses.
 5. Allow users to select available dates and time slots.
 6. Enable secure online payment through Razorpay.
 7. Verify payments on the backend.
 8. Confirm successful bookings automatically.
 9. Notify both the user and administrator about successful payments.
 10. Automatically send reminders before the scheduled webinar.
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3. Business Problem

Traditional webinar registration processes often require users to communicate manually with administrators, make payments separately, and depend on manual reminders.

This can result in:

- Manual booking management
- Difficulty tracking available seats
- Payment verification delays
- Missed webinar sessions
- Manual reminder work
- Lack of centralized booking information
- Difficulty monitoring payments and users

Skill Cortex aims to automate this entire process through a single platform.

4. Project Scope

4.1 In Scope

The following functionality is included in Version 1:

- Landing page/dashboard
- User registration
- User login
- User profile
- Department selection
- Webinar/course listing
- Webinar details
- Date and time slot selection
- Slot availability
- Booking management
- Razorpay payment integration
- Payment verification
- Booking confirmation
- Email notifications
- SMS notifications
- Automated reminders
- User booking history
- Admin dashboard
- Department management
- Webinar management
- Slot management
- User management
- Booking management
- Payment monitoring
- Notification tracking

4.2 Out of Scope

The following are not part of V1:

- AI-based skill recommendations
- AI resume analysis
- AI career recommendations
- AI interview preparation
- Skill-gap analysis
- Learning recommendation engine
- Advanced analytics
- AI agents

These can be introduced in future versions.

5. Stakeholders

Stakeholder	Responsibility
User	Browse webinars, book slots and make payments
Admin	Manage webinars, slots, users and payments
System Administrator	Maintain the application and infrastructure
Payment Provider	Process online payments
Notification Provider	Deliver email/SMS notifications

6. User Roles

6.1 User

A registered user can:

- Access the platform
- Select a department
- Browse webinars
- View webinar details
- Select a date/time slot
- Make payment
- View booking status
- View payment details
- Receive confirmation
- Receive reminders

6.2 Admin

An administrator can:

- Log in to the admin dashboard
 - Manage departments
 - Manage webinars
 - Create available slots
 - Manage users
 - View bookings
 - View payments
 - Monitor notification status
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7. Overall Business Workflow

The complete business workflow is:

Landing Page / Dashboard

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Register / Login

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Select Department

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Select Webinar / Course

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Select Date & Time Slot

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Confirm Booking

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Razorpay Payment

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Backend Payment Verification

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Booking Confirmation

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User + Admin Email/SMS Notification

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Automated Daily Reminder

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Webinar Attendance

8. Functional Requirements

FR-01: Landing Page

The system shall provide a public landing page for Skill Cortex.

The landing page should display:

- Skill Cortex branding
- Introduction/about section
- Available webinars/courses
- Departments
- Course highlights
- Benefits
- Login button
- Registration button
- Contact information

Users should be able to explore basic course information before logging in.

FR-02: User Registration

The system shall allow new users to register.

Required information may include:

- Full name
- Email address
- Mobile number
- Password

- Department

The system shall validate the submitted information before creating the account.

FR-03: User Login

Registered users shall be able to log in using their credentials.

The system shall authenticate the user and provide access to the user dashboard.

FR-04: Department Selection

Users shall be able to select their department.

Example departments:

- Computer Science and Engineering
- Artificial Intelligence / Machine Learning
- Data Science

The administrator shall be able to add or modify departments.

FR-05: Webinar/Course Management

The system shall display available webinars and courses.

Each webinar should contain:

- Webinar title
- Description
- Department
- Price
- Duration
- Instructor information, if applicable
- Available slots
- Status

Example:

Python Programming Webinar

Department: CSE / AI-ML / Data Science

Price: ₹999

FR-06: Slot Management

Administrators shall be able to create webinar slots.

Each slot shall contain:

- Webinar
- Date
- Start time
- End time
- Maximum capacity
- Available seats
- Slot status

The system shall prevent users from booking unavailable or fully occupied slots.

FR-07: Booking

Users shall be able to book a selected webinar and slot.

The system shall store:

- User
- Department
- Webinar
- Slot
- Booking date
- Booking status

A booking may initially have a **PENDING** status until payment is successfully verified.

FR-08: Razorpay Payment

The platform shall integrate Razorpay for online payments.

Payment workflow:

1. User selects a webinar.
2. User selects a slot.
3. System creates a booking/payment request.
4. Backend creates a Razorpay order.
5. User completes payment.
6. Razorpay returns payment information.
7. Backend verifies the payment.
8. Payment is marked as successful.
9. Booking is confirmed.

The system shall not rely solely on frontend payment success.

FR-09: Payment Management

The system shall maintain payment information including:

- Booking ID
- Razorpay order ID
- Razorpay payment ID
- Amount
- Payment status
- Payment date/time

Possible payment statuses:

- PENDING
 - PAID
 - FAILED
 - REFUNDED
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FR-10: User Notification

After successful payment verification, the system shall notify the user.

The notification should contain:

- User name
- Webinar name
- Date
- Time
- Amount paid
- Booking confirmation
- Payment reference

Notifications shall support:

- Email
 - SMS
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FR-11: Admin Notification

After successful payment verification, the administrator shall receive a notification.

The notification should contain:

- User name
- User contact information
- Webinar

- Date/time slot
 - Amount paid
 - Payment status
 - Transaction/payment reference
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FR-12: Automated Reminder

The system shall automatically send reminders to users based on their booked webinar slot.

Example:

3 days before:

"Your Python webinar is in 3 days."

1 day before:

"Your Python webinar is tomorrow at 10:00 AM."

Day of webinar:

"Reminder: Your Python webinar starts today at 10:00 AM."

The reminder schedule should be configurable.

FR-13: User Dashboard

The user dashboard shall display:

- Profile information
 - Selected department
 - Upcoming webinars
 - Previous webinars
 - Booking status
 - Payment status
 - Webinar date/time
 - Notification information
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FR-14: Admin Dashboard

The admin dashboard shall provide an overview of:

- Total users
- Total webinars
- Upcoming slots
- Total bookings
- Successful payments
- Pending payments
- Failed payments

- Upcoming webinars
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9. Business Rules

BR-01

Only registered/authenticated users can make webinar bookings.

BR-02

A user must select a department before selecting department-specific webinars.

BR-03

A user can book only an available slot.

BR-04

A slot cannot exceed its configured capacity.

BR-05

A booking shall not be considered confirmed until payment is successfully verified.

BR-06

Payment verification must happen on the backend.

BR-07

Confirmation notifications should be triggered only after successful payment verification.

BR-08

Reminders should be sent only for valid/confirmed paid bookings.

BR-09

The reminder system should prevent duplicate reminders.

BR-10

Administrators can create, modify, activate, or deactivate webinars and slots.

10. Non-Functional Requirements

Security

The system should:

- Hash user passwords securely.
- Use authenticated API endpoints.
- Protect admin routes.
- Store secrets in environment variables.
- Validate user input.
- Verify Razorpay payments server-side.
- Use HTTPS in production.

Performance

The system should:

- Load major pages quickly.
- Support multiple concurrent users.
- Avoid double-booking slots.
- Process payment callbacks reliably.

Reliability

The system should:

- Maintain accurate payment records.
- Maintain booking history.
- Track notification status.
- Handle failed payment attempts gracefully.
- Prevent duplicate reminders.

Scalability

The architecture should allow future additions such as:

- AI recommendations
 - Skill analysis
 - More departments
 - More courses
 - Multiple instructors
 - Advanced analytics
 - Mobile application
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11. Data Requirements

The core entities are:

User

- User ID
- Name
- Email
- Phone
- Password
- Department
- Created date

Department

- Department ID
- Department name
- Description
- Status

Webinar

- Webinar ID
- Title
- Description
- Department
- Price
- Duration
- Status

Slot

- Slot ID
- Webinar ID
- Date
- Start time
- End time
- Capacity
- Available seats
- Status

Booking

- Booking ID
- User ID
- Webinar ID
- Slot ID
- Booking status
- Created date

Payment

- Payment ID
- Booking ID
- Razorpay order ID
- Razorpay payment ID
- Amount
- Payment status
- Paid date

Notification

- Notification ID
- User ID
- Booking ID
- Notification type
- Channel
- Message
- Status
- Sent date

12. Notification Requirements

The notification service should support two channels:

Email

and

SMS

Notification types:

Notification	Recipient	Trigger
Registration	User	Successful registration
Booking confirmation	User	Verified payment
Payment notification	Admin	Verified payment
Reminder	User	Upcoming webinar
Cancellation	User	Booking cancellation, if supported

13. Admin Management Workflow

Admin logs in

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Admin dashboard

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Manage departments

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Create webinar

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Set webinar price

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Create available slots

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Monitor bookings

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Monitor payments

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Monitor notifications

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Manage upcoming webinars

14. User Experience Requirements

The platform should provide a simple and clear journey:

Discover → Register → Select → Book → Pay → Confirm → Remember → Attend

The user should always be able to clearly see:

- What webinar they selected
- Which date/time they selected

- How much they need to pay
 - Whether payment succeeded
 - Whether their booking is confirmed
 - When the webinar is scheduled
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15. Success Criteria

Skill Cortex V1 will be considered successful when:

1. Users can access the landing page.
 2. Users can register and log in.
 3. Users can select their department.
 4. Users can browse available webinars.
 5. Users can select available slots.
 6. Users can complete Razorpay payments.
 7. The backend successfully verifies payments.
 8. Confirmed bookings are stored correctly.
 9. Users receive payment/booking confirmation.
 10. Admin receives payment notification.
 11. Users receive automated webinar reminders.
 12. Admin can manage webinars and slots.
 13. The system prevents unavailable-slot bookings and duplicate reminders.
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16. Future Enhancements

Future versions can extend Skill Cortex into an AI-powered learning platform.

Potential V2 features include:

- AI skill assessment
 - Resume analysis
 - Skill-gap detection
 - Personalized webinar recommendations
 - AI career roadmap
 - AI interview preparation
 - Job-role matching
 - Learning progress tracking
 - AI-powered course recommendations
 - Agentic AI learning assistant
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17. Final V1 Business Flow

Landing Page

- **User Registration/Login**
 - **Department Selection**
 - **Webinar Selection**
 - **Date & Time Slot Selection**
 - **Booking**
 - **Razorpay Payment**
 - **Backend Verification**
 - **Booking Confirmation**
 - **User Notification**
 - **Admin Notification**
 - **Automated Reminder**
 - **Webinar Attendance**
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Conclusion

Skill Cortex V1 will provide a complete digital workflow for managing paid webinars and courses. It will reduce manual registration and payment management, improve the user booking experience, provide administrators with centralized control, and ensure users receive timely reminders for their scheduled sessions.

The V1 architecture will also provide a strong foundation for adding AI-powered skill intelligence and personalized learning features in future releases.